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Capacity Achieving Cyclic Codes

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A family of codes is said to be **capacity achieving** on the q -ary erasure channel if the sequence of rates converges to some $r \in (0, 1)$ and for all error probabilities, $p \in (0, 1 - r)$, the probability of a decoding failure converges to zero as $n \rightarrow \infty$.

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- This definition is equivalent to Shannon's, but focuses on the convergence of the probability of decoding failure as the channel capacity channels converges to r rather than as the rate converges to r on a fixed channel.

Capacity Achieving Codes (continued)

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Remark: The capacity achieving property depends on both the channel and the decoder.

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Remark: The capacity achieving property depends on both the channel and the decoder.

- In this talk, the channel will be a q -ary erasure channel

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Remark: The capacity achieving property depends on both the channel and the decoder.

- In this talk, the channel will be a q -ary erasure channel
- All of the results are with respect to symbolwise *Maximal a Posteriori* decoding (Symbol-MAP or bit-MAP for binary codes)

Capacity Achieving Codes (continued)

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Remark: The capacity achieving property depends on both the channel and the decoder.

- In this talk, the channel will be a q -ary erasure channel
- All of the results are with respect to symbolwise *Maximal a Posteriori* decoding (Symbol-MAP or bit-MAP for binary codes)
- Symbol-MAP decoding is very robust, but usually very inefficient. Hence, this is a “best case” for a family to be capacity achieving.

Code Permutation Groups

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Code Permutations: Given a code $\mathcal{C}$ of length n , the automorphism group of $\mathcal{C}$ is given by

$$\text{Perm}(\mathcal{C}) = \{P \in S_n \mid \forall c \in \mathcal{C}, cP \in \mathcal{C}\}.$$

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Although $\text{Perm}(\mathcal{C})$ is defined based on the action on the codewords, we consider the action of $\text{Perm}(\mathcal{C})$ on the set $\{0, \dots, n-1\}$, the position coordinates of $\mathcal{C}$.

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Although $\text{Perm}(\mathcal{C})$ is defined based on the action on the codewords, we consider the action of $\text{Perm}(\mathcal{C})$ on the set $\{0, \dots, n-1\}$, the position coordinates of $\mathcal{C}$.

Define $G_0 \leq \text{Perm}(\mathcal{C})$ to be the stabilizer of the 0 coordinate of codewords in $\mathcal{C}$.

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Define $G_0 \leq \text{Perm}(\mathcal{C})$ to be the stabilizer of the 0 coordinate of codewords in $\mathcal{C}$.

- If for every $i, j \in \{0, \dots, n-1\}$ there is some $P \in \text{Perm}(\mathcal{C})$ such that $P(i) = j$ then $\text{Perm}(\mathcal{C})$ is **transitive**.

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Define $G_0 \leq \text{Perm}(\mathcal{C})$ to be the stabilizer of the 0 coordinate of codewords in $\mathcal{C}$.

- If for every $i, j \in \{0, \dots, n-1\}$ there is some $P \in \text{Perm}(\mathcal{C})$ such that $P(i) = j$ then $\text{Perm}(\mathcal{C})$ is **transitive**.
- If G_0 is transitive on the set $\{1, \dots, n-1\}$ then $\text{Perm}(\mathcal{C})$ is **doubly transitive**.

Permutations and Capacity Achieving Codes

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Let $\{\mathcal{C}_n\}$ be a sequence of codes with increasing blocklengths, where the rates satisfy $r_n \rightarrow r \in (0, 1)$.

Theorem 1 (Kudekar et al 2016): If $\text{Perm}(\mathcal{C}_n)$ is doubly transitive for each n then $\{\mathcal{C}_n\}$ is capacity achieving on the binary erasure channel under bit-MAP decoding.

Example: This implies that Reed-Muller codes are capacity achieving on the binary erasure channel.

Permutations and Capacity Achieving Codes

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Let m_n denote the smallest orbit size of any index under the action of G_0 .

Theorem 2 (Kumar, Calderbank, and Pfister 2016): If $\text{Perm}(\mathcal{C}_n)$ is transitive for each n and $m_n \rightarrow \infty$ then $\{\mathcal{C}_n\}$ is capacity achieving on the q -ary erasure channel under bit-MAP decoding.

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Let m_n denote the smallest orbit size of any index under the action of G_0 .

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- We refer to this condition as the “Transitivity + Orbit” condition.

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Cyclic Code: An $[n, k]$ linear code $\mathcal{C}$ over $\mathbb{F}_q$ is called cyclic, if for every codeword $c = [c_1, c_2, \dots, c_{n-1}, c_n]$ the vector $[c_n, c_1, c_2, \dots, c_{n-1}]$ is also a codeword of $\mathcal{C}$.

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Alternatively:

Cyclic Code: Let $g(x) \in \mathbb{F}_q[x]$ be a factor of $x^n - 1$. The cyclic code defined by $g(x)$ is the ideal $(g(x))$ in the ring $\mathbb{F}_q[x]/(x^n - 1)$. for a polynomial $a_0 + a_1x + \dots + a_nx^n$

Key Permutations of Cyclic Codes

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Let $\mathcal{C}$ is an $[n, k]_q$ linear code. If n is relatively prime to q then:

- The permutation $\pi : i \mapsto i + 1 \pmod n$ is contained in $\text{Perm}(\mathcal{C})$

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- The permutation $\phi : i \mapsto qi \pmod n$ is contained in $\text{Perm}(\mathcal{C})$

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Let $\mathcal{C}$ is an $[n, k]_q$ linear code. If n is relatively prime to q then:

- The permutation $\pi : i \mapsto i + 1 \pmod n$ is contained in $\text{Perm}(\mathcal{C})$
- The permutation $\phi : i \mapsto qi \pmod n$ is contained in $\text{Perm}(\mathcal{C})$

The permutation π ensures that $\text{Perm}(\mathcal{C})$ is transitive, while ϕ (called the Frobenius automorphism) is a permutation which stabilizes the first position.

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An analysis of the orbits induced by ϕ will be central to our discussion. So, we define the following:

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An analysis of the orbits induced by ϕ will be central to our discussion. So, we define the following:

- $\phi_n : i \mapsto qi \pmod n$ (this will be useful in discussing ϕ with respect to different lengths)

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An analysis of the orbits induced by ϕ will be central to our discussion. So, we define the following:

- $\phi_n : i \mapsto qi \pmod n$ (this will be useful in discussing ϕ with respect to different lengths)
- The order of ϕ_n is also the size of the largest cyclotomic coset which in the binary case is denoted $\text{ord}_n(2)$.

When do Cyclic Codes Achieve Capacity

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Theorem 3 (Kumar, Calderbank, and Pfister 2016): Let $\{\mathcal{C}_n\}$ be a sequence of $\mathbb{F}_q$ cyclic codes with rates $r_n \rightarrow r \in (0, 1)$ and increasing blocklengths N_n .

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Theorem 3 (Kumar, Calderbank, and Pfister 2016): Let $\{\mathcal{C}_n\}$ be a sequence of $\mathbb{F}_q$ cyclic codes with rates $r_n \rightarrow r \in (0, 1)$ and increasing blocklengths N_n . Suppose N_n divides $q^{t_n} - 1$ and is relatively prime to $q - 1$ and let s_n be the smallest prime divisor of t_n .

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Important Lemmas

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Lemma 1: Let $n = p_1^{a_1} \dots p_k^{a_k}$ where p_1 through p_k are distinct primes and let $q = p^m$ such that $\gcd(n, q) = 1$. Then the smallest orbit induced by ϕ_n has the same length as the least orbit induced by ϕ_{p_i} for any p_i .

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Lemma 2: Let p be a prime and q a power of a different prime. Suppose that t is the largest integer such that $q^t < p$. Then for any $a \in [p-1]$ the order of a under the Frobenius map is at least t .

Theorem 4

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Theorem 4: Let $\{\mathcal{C}_n\}$ be a family of q -ary cyclic codes with increasing block lengths N_n and rates $r_n \rightarrow r \in (0, 1)$. Let p_n be the least prime divisor of N_n . If $p_n \rightarrow \infty$ then $\{\mathcal{C}_n\}$ is capacity achieving on the q -ary erasure channel under MAP-decoding.

Proof of Theorem 4

Proof:

Transitivity is free!

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Proof of Theorem 4

Proof:

Transitivity is free!

Let p_n be the least prime divisor of N_n and t_n the largest integer such that $q^{t_n} < p_n$.

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Proof of Theorem 4

Proof:

Transitivity is free!

Let p_n be the least prime divisor of N_n and t_n the largest integer such that $q^{t_n} < p_n$.

By Lemma 1, the smallest orbit size for in N_n is that of the least orbit induced on one of the prime divisors of N_n .

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By Lemma 1, the smallest orbit size for in N_n is that of the least orbit induced on one of the prime divisors of N_n .

For each prime dividing N_n the t defined in Lemma 2 is a lower bound on the size of the orbit induced by that prime. Each of which is bounded below by t_n .

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For each prime dividing N_n the t defined in Lemma 2 is a lower bound on the size of the orbit induced by that prime. Each of which is bounded below by t_n .

By assumption, $p_n \rightarrow \infty$ and q is fixed, meaning that $t_n \rightarrow \infty$. Therefore $\{\mathcal{C}_n\}$ satisfies the Transitivity + Orbit condition and is capacity achieving on the q -ary erasure channel under MAP-decoding. □

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Remark:

- Theorem 4 is easier to apply since it does not require exponentiation.

Remark:

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- Theorem 4 applies to a broader range of codes and implies Theorem 3.

Remarks on Theorem 4 Continued

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- For any sequence of lengths N_n which does not have an increasing minimum prime divisor, we can construct a family of codes with rate $r \in (0, 1)$ and which are not capacity achieving.

Remarks on Theorem 4 Continued

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- For any sequence of lengths N_n which does not have an increasing minimum prime divisor, we can construct a family of codes with rate $r \in (0, 1)$ and which are not capacity achieving.
- In 2023, Natajaran and Krishnan showed the existence of capacity achieving cyclic codes whose block lengths have a fixed set of prime factors.

Remarks on Theorem 4 Continued

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- For any sequence of lengths N_n which does not have an increasing minimum prime divisor, we can construct a family of codes with rate $r \in (0, 1)$ and which are not capacity achieving.
- In 2023, Natajaran and Krishnan showed the existence of capacity achieving cyclic codes whose block lengths have a fixed set of prime factors.
- In practice, showing that an $\{N_n\}$ admits a family of codes with $r \in (0, 1)$ is as difficult as showing that it satisfies the increasing minimal prime divisor condition.

Existence of Binary Cyclic Codes of Rate r

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Lemma 3: Let $\{N_n\}$ be a sequence of increasing odd integers. If $\left\{ \frac{\text{ord}_{N_n}(2)}{N_n} \right\} \rightarrow 0$ then for any rational $r \in (0, 1)$ there exists a family of cyclic codes $\{\mathcal{C}_n\}$ with lengths $\{N_n\}$ and rates converging to r .

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Remark:

- The proof of Lemma 3 depends on a bin packing argument to form a generating polynomial that defines a code with approximately the correct rate.

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Remark:

- The proof of Lemma 3 depends on a bin packing argument to form a generating polynomial that defines a code with approximately the correct rate.
- If $\{N_n\}$ satisfies both Lemma 3 and the Transitivity + Orbit condition then we may construct capacity achieving codes with any rate.

Zsigmondy Theorem

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A prime p is a **primitive prime** divisor of s_m if $p \mid s_m$ and $p \nmid s_k$ for any $k < m$.

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A prime p is a **primitive prime** divisor of s_m if $p \mid s_m$ and $p \nmid s_k$ for any $k < m$.

Theorem (Zsigmondy 1982): If $1 \leq b < a$ such that $\gcd(a, b) = 1$ then each term in the sequence $\{a^n - b^n\}$ has at least one primitive prime factor except in some cases where $n \leq 6$.

Zsigmondy Theorem

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Remark: In particular, $2^n - 1$ has a primitive prime divisor for $n > 6$ which will allow us to control $\text{ord}_n(2)$.

Suitable Block Lengths

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For any sequence $\{m_n\}$ of increasing composite natural numbers we may find a sequence $\{N_n\}$ by:

- Define $\{b_n\}$ to be a sequence of proper divisors of $\{m_n\}$ such that $b_n \geq \sqrt{m_n}$

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For any sequence $\{m_n\}$ of increasing composite natural numbers we may find a sequence $\{N_n\}$ by:

- Define $\{b_n\}$ to be a sequence of proper divisors of $\{m_n\}$ such that $b_n \geq \sqrt{m_n}$
- Define $A_n := \{a \mid a \text{ is prime, } a < \lg(m_n)\}$

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- Define $\{b_n\}$ to be a sequence of proper divisors of $\{m_n\}$ such that $b_n \geq \sqrt{m_n}$
- Define $A_n := \{a \mid a \text{ is prime, } a < \lg(m_n)\}$
- Let $\{N_n\}$ be a sequence of integers such that N_n is a factor of $2^{m_n} - 1$ which is divisible by at least one primitive prime of $2^{m_n} - 1$, one primitive prime of $2^{b_n} - 1$, and is relatively prime to a for all $a \in A_n$.

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For any sequence $\{m_n\}$ of increasing composite natural numbers we may find a sequence $\{N_n\}$ by:

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- Define $A_n := \{a \mid a \text{ is prime, } a < \lg(m_n)\}$
- Let $\{N_n\}$ be a sequence of integers such that N_n is a factor of $2^{m_n} - 1$ which is divisible by at least one primitive prime of $2^{m_n} - 1$, one primitive prime of $2^{b_n} - 1$, and is relatively prime to a for all $a \in A_n$.

Remark: It is always possible to make such a construction.

Theorem 5

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Theorem 5: If $\{N_n\}$ is constructed as described, then for any rational $r \in (0, 1)$ there exists a family of cyclic codes $\{\mathcal{C}_n\}$ with lengths $\{N_n\}$ such that $\{\mathcal{C}_n\}$ has rates converging to r and is capacity achieving on the BEC under MAP decoding.

Proof Idea:

Being relatively prime to the elements of A_n ensures that the smallest prime dividing N_n is increasing.

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Proof Idea:

Being relatively prime to the elements of A_n ensures that the smallest prime dividing N_n is increasing.

By construction, we know that $N_n \geq m_n \sqrt{m_n}$ meaning $\lim_{n \rightarrow \infty} \frac{m_n}{N_n} = 0$. So, Lemma 3 applies and there is a sequence of binary cyclic codes with lengths N_n and rates converging to r .

Theorem 5

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By construction, we know that $N_n \geq m_n \sqrt{m_n}$ meaning $\lim_{n \rightarrow \infty} \frac{m_n}{N_n} = 0$. So, Lemma 3 applies and there is a sequence of binary cyclic codes with lengths N_n and rates converging to r . Therefore, Theorem 4 applies and the sequence of codes is capacity achieving. □

Concluding Remarks

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- Theorem 5 shows that there are certain sequences of lengths which allow for capacity achieving codes at any rate r .

Concluding Remarks

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- Theorem 5 shows that there are certain sequences of lengths which allow for capacity achieving codes at any rate r .
- Theorem 5 also serves to demonstrate that Theorem 4 is a strict improvement over Theorem 3.

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- Theorem 5 shows that there are certain sequences of lengths which allow for capacity achieving codes at any rate r .
- Theorem 5 also serves to demonstrate that Theorem 4 is a strict improvement over Theorem 3.
- All of the cyclic code work from Theorem 3 to Theorem 5 rests on the analysis of only two permutations so it is likely that additional conditions can be developed if more is known of the code structure.

Key References

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References

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